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Foreground-Background Separation through Concept Distillation from Generative Image Foundation Models

Mischa Dombrowski[1] Hadrien Reynaud[2] Matthew Baugh[2] Bernhard Kainz[1,2]
[1]Friedrich–Alexander–Universität Erlangen–Nürnberg [2]Imperial College London
mischa.dombrowski@fau.de

Abstract

Curating datasets for object segmentation is a difficult task. With the advent of large-scale pre-trained generative models, conditional image generation has been given a significant boost in result quality and ease of use. In this paper, we present a novel method that enables the generation of general foreground-background segmentation models from simple textual descriptions, without requiring segmentation labels. We leverage and explore pre-trained latent diffusion models, to automatically generate weak segmentation masks for concepts and objects. The masks are then used to fine-tune the diffusion model on an inpainting task, which enables fine-grained removal of the object, while at the same time providing a synthetic foreground and background dataset. We demonstrate that using this method beats previous methods in both discriminative and generative performance and closes the gap with fully supervised training while requiring no pixel-wise object labels. We show results on the task of segmenting four different objects (humans, dogs, cars, birds) and a use case scenario in medical image analysis. The code is available at https://github.com/MischaD/fobadiffusion.

1. Introduction

Supervised pretraining, *e.g.*, with ImageNet[16], has demonstrated reduced training times and boosted performance. This gave rise to models that could be trained once over large amounts of data before being adapted to specialised tasks, such as image recognition, object detection, image segmentation[52], and medical image analysis[50]. The recent development of self-supervision techniques and their ability to learn without manual labels led to much larger scale training datasets [9] and to the creation of foundation models [6]. At present, the use of pre-trained models for a wide range of diverse downstream tasks defines a very active and intriguing area of research.

Large scale foundation models are already established in natural language processing, with most of them being based on the Transformer architecture [57, 17, 7]. A crucial part of this architecture are cross- and self-attention layers, which compute interpretable importance weightings [39, 12].

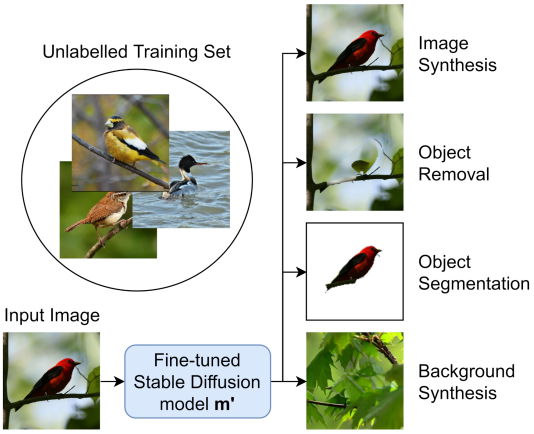


Figure 1. High level overview of our proposed method: Without needing a single labelled image, our method is able to generate foreground, background, and segmentation masks for any concept that is known to a text-to-image generative network.

Diffusion models are based on the U-Net architecture [45] with additional attention layers [37] to condition on textual prompts. Therefore, we can extract inherently interpretable pixel importance scores from conditioning on textual prompts. Furthermore, the reverse diffusion process teaches the U-Net to successively remove noise from images, starting from pure Gaussian noise. In the early steps of this process, where the images resemble pure noise, the texture is non-existent, and the model only learns structures.

Recently, latent diffusion models have emerged as state-of-the-art generative models for the task of text-to-image generation [18, 44, 41]. However, training such models requires a significant amount of CO_2-intensive resources and until recently, pre-trained model weights have not been publicly available. Rombach et al. were the first to published their weights and model architecture [44], which facilitated the development of numerous derived applica-

tions [60, 15, 53] and established this model as the foundation model for tasks that require generalised representations of concepts in images. State-of-the-art latent diffusion models are able to generate high resolution images of a vast amount of different objects, suggesting that a highly expressive latent representation of the data has been learned.

We hypothesize that we can leverage these learned latent representations for our own downstream tasks of zero-shot foreground-background generation. Using a generative latent diffusion foundation model, we are able to extract a weak segmentation mask around an arbitrary object by computing the importance maps based on the textual input prompts. Weak segmentation masks have been shown to be an effective prior for segmentation models, given that enough training samples are available [34, 40]. We then use these preliminary masks to fine-tune a latent diffusion model on (1) generating new images from this dataset, as well as (2) inpainting regions where the object is not present according to our preliminary masks. The resulting model is able to perform full-image synthesis, as well as foreground, background and mask generation, as summarized in Fig. 1. A segmentation model trained using these masks can then achieve a level of performance that is close to direct supervision, despite not requiring manual segmentation masks at any point in the pipeline. This suggests that labour-intensive ground-truth image annotation workflows could become obsolete in the future, and be replaced by concept distillation from generative foundation models.
Our main contributions are:

- We propose a self-supervised, hyperparameter-free, approach for dataset-independent foreground-background segmentation, based on latent diffusion models, capable of synthesizing foreground, background, and segmentation masks.
- We describe a general framework to extract importance scores obtained from pretrained diffusion models and detail how to use them to improve segmentation performance.
- We verify the feasibility of our method on a set of four different foreground background segmentation tasks, spanning humans, birds, dogs, and cars and show that our method achieves results close to supervised methods while being trained without direct supervision.
- We experiment with the extension of our method to domain-adapted diffusion models by showing promising results on a medical segmentation task.

2. Related Work

Semantic Segmentation refers to the identification of high level concepts in an image, which enable their extraction from the image. [33, 36] introduced the use of fully convolutional networks for this task, which superseded previous shallow feature classification approaches [20]. Currently, the most common segmentation network architectures are designed as encoder-decoder pairs, as it enforces an information bottleneck that facilitates generalisation. The encoder provides meaningful low dimensional representations and the decoder reconstructs high-resolution segmentation maps [10, 45, 33, 36]. Recent approaches maximise the use of multi-scale information with multi-scale attention [56], squeeze-and-attention [63], and Transformers [62]. The final pixel classification operation, which creates the segmentation mask, is performed through multinomial logistic regression. These methods require large amounts of manually labelled training samples, which can be labour-intensive and expensive.

Weakly Supervised Semantic Segmentation partially mitigates that by learning from weak global labels, such as image-wise class labels, and perform rough semantic-segmentation tasks. These approaches often leverage the learned representations in intermediate layers through attention maps [26] or saliency maps [31] extraction. These representations are learned in a supervised manner through a classification task, as opposed to our approach that leverages an even weaker signal: free-form text-embeddings. Furthermore, these methods cannot generate images and are, therefore, not suitable for foreground-background synthesis.

Foreground background separation is a segmentation task where the goal is to apply binary classification over all the pixels of an image to separate the object of interest, the foreground, from the contextual background. For example, in video analysis tasks, the background is often defined as parts of scenes that are at rest [51]. More recently, decomposing individual images into potential foreground and background layers became an intriguing research topic [25]. Since the foreground-background separation factors are not known a priori, many related works formulate the problem as a category-agnostic unsupervised segmentation approach. While deep neural networks can learn pixel clustering in an unsupervised way [25, 29, 38], it often leads to inferior performance when compared to fully supervised methods. Other works also tried to learn image segmentation from generative models, for example via direct sampling from the training distribution with cut-and-paste [43], image combinations via styleGAN [1], erasing and redrawing [11], and through inpainting [49].

The work closest related to ours is [61] who employ layered generative adversarial networks (GANs) to generate distinct images for the foreground and background. As opposed to [61], we are not relying on the unknown structure of the latent manifold to separate the embedding codes that represent foreground and background, but instead follow the directly interpretable paradigm 'textual concept description' $\rightarrow$ 'segmentation model'. Both approaches in-

troduce a similar bias, ours through the textual concept description and them through object-specific models, making our method similarly conditioned but more flexible.

Diffusion models are generative methods. Generative modeling has always been an important task in deep learning. Recently, Diffusion models gained a lot of attention thanks to the impressive results achieved by [42, 48]. Based on [22, 55], these models currently define the state-of-the-art in conditional image generation, and have been extended towards text-to-image models such as unCLIP [41] and Stable Diffusion [44]. Part of that success is due to the efficacy of straightforward extensions to diffusion models, such as classifier-free guidance [23]. Current research about diffusion models focuses mostly on inpainting [14, 13, 44]. Depending on the task, it might be easier to learn the object itself and then train a diffusion model on the task of inpainting regions where the object is not present. However, we focus on the scenario where we have the object itself present and try to remove it.

Shortly after latent diffusion models showed superiority in terms of image sample quality compared to GANs [18], they were conditioned on the description of concepts [42]. These models became widely available online, including on commodity hardware after the computationally expensive denoising process was accelerated on a fundamental level [32]. Direct translation of text into object-centric representations has been attempted but rather in the context of attention editing [21], subject specific image generation [47], and textual concept-refined image-to-image translation [59]. Textual inversion, where examples of concepts are provided to a diffusion model as text and image tuples to teach the model a new concept [19], operates on the same fundamental input level as our approach. However, it is not able to provide object segmentation masks or concept-specific pixel importance scores. Recent advances on diffusion models have led to a spike in research around how latent features can be extracted [21, 2] however, their potential to be used for zero-shot segmentation has yet to be explored.

3. Method

Our approach is summarized in Fig. 2. At a high level, we first use the attention maps from a pretrained latent diffusion model to compute coarse foreground segmentations (preliminary masks). We use these masks to fine-tune the diffusion model to be able to remove the object from the image, replacing it with background information, whilst also being able to generate new samples from the true distribution $p(\mathbf{x})$ of the dataset. By comparing the original images to those with the background inpainted over the foreground, we are then able to produce much more refined masks. We can also then use the fine-tuned model to sample an arbitrary number of new images, including only the foreground, only the background, and the masks separating them. The only assumptions this method uses is that we have access to a dataset of images where the chosen object is always present, and that the foundation model we use has learned a meaningful representation of the object we want to identify. In this context, objects can describe concepts like birds, cars, dogs, humans, but also parts of objects such as arms or legs.

We start from the latent diffusion model (LDM) [44], a foundation model pretrained on the task of text-to-image generation, denoted as m. It generates images by sampling gaussian noise and iteratively denoising them in T diffusion steps. Throughout this work, we keep the default value from [44] of $T = 50$ steps.

Let $\mathbf{D}$ denote a dataset where all the images contain our desired object. Formally, we aim to generate unsupervised segmentation masks of the original dataset $\mathbf{D_m}$ as well as a synthetic dataset that contains synthetic images $\mathbf{D'}$, masks $\mathbf{D'_m}$, foreground $\mathbf{D'_f}$, and background $\mathbf{D'_b}$ denoted as set $\mathbf{D'_s} = \{\mathbf{D'}, \mathbf{D'_m}, \mathbf{D'_f}, \mathbf{D'_b}\}$.

Preliminary Masks: The LDM was trained on paired text-image data. Thus, the output is conditioned on a text input, which we have to carefully choose to generate our initial attention maps. In practice, self- and cross-attention work well for conditioning on different inputs, especially across different modalities [57]. As input, we propose the prompt "*a photo of a* {*object*}" where "object" is a high level description of our foreground object (*e.g.* "bird").

The first step in computing the preliminary masks is to leverage the raw attention maps computed in every cross-attention layer of the U-Net architecture. We decide against cherry-picking different attention layers for different tasks to remain task-agnostic, but would like to point out that this could lead to task-specific improvements as an hyperparameter option.

Let z_0 denote the latent space representation of some input image x. In each layer, attention is computed as:

$$\text{Attention}(Q, K, V) = \psi(Q_l, K_l^T) \cdot V_l \tag{1}$$

with the attention probabilities $\psi(Q_l, K_l^T)$ defined as:

$$\psi_{\mathbf{z_{t,l}}}(Q_l, K_l^T) = \text{softmax}(\frac{QK^T}{\sqrt{d}}) \tag{2}$$

with $Q = W_Q^{(i,l)} \cdot \phi(\mathbf{z_{t,l}})$, $K = W_K^{(i,l)} \cdot \tau_\theta(y)$, and $V = W_V^{(i,l)} \cdot \tau_\theta(y)$ denoting the learnable projection matrices according to [57], $\phi(\mathbf{z_{t,l}})$ the latent code of the l-th U-Net layer in the t-th reverse diffusion step, and $\tau_\theta(y)$ the learned latent representation for the textual input prompt [44]. $z_{t,l}$ denotes the latent representation of the stable diffusion model in layer l, conditioned at diffusion t.

The next step is to compute the mean attention maps $\hat{M}$

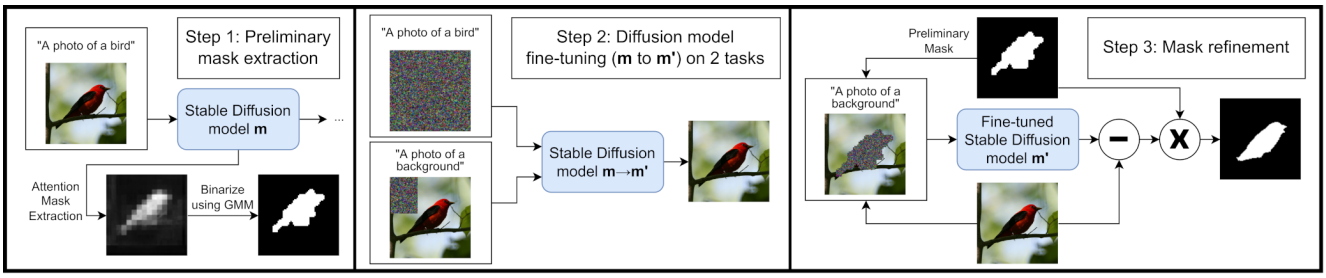


Figure 2. Overview of our model pipeline for self-supervised foreground-background segmentation.

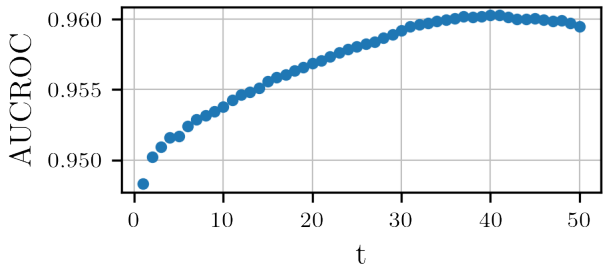


Figure 3. T_0 vs AUC-ROC on CUB. Incorporating more reverse diffusion steps into the attention computation improves the AUC-ROC against the groundtruth only up to roughly $T_0 = 40$.

as expectations over multiple repeated diffusion steps according to

$$\hat{M} = \sum_{t=1}^{T_0} \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z_t}|\mathbf{z_{t+1}},\mathbf{z_0})}[\sum_{l} \psi_{\mathbf{z_{t,l}}}(Q_l, K_l^T)]. \quad (3)$$

We empirically show in our supplementary material that this can be simplified to performing single reverse diffusion steps which leads to a simplified importance score

$$\hat{M} = \sum_{t=1}^{T_0} \sum_{l} \psi_{\mathbf{z_{t,l}}}(Q_l, K_l^T) \quad (4)$$

Figure 3 shows AUCROC of using $\hat{M}$ to identify the foreground as a function of the number of reverse steps. There is a clear improvement visible computing the mean over up to roughly $T_0 = 40$ reverse diffusion steps. Including higher diffusion steps seems to deteriorate the accuracy of ϕ. This is because for higher values of t the input images approach pure Gaussian noise. However, for medium values of t the input and output images already approximate the basic structure of the final objects. Other details, like texture, appear at later stages but do not bear any valuable information for this use case. This observation could mean that for time critical applications, starting from lower t should suffice.

The third and final step in retrieving the preliminary masks is to binarize the mean attention maps. To do so, we take advantage of the observation that all attention maps resemble a bimodal distribution, with one mode at a low value for the non-object pixels and one mode at a high value for the object pixels. Hence, we model absolute values of the instance-wise attention scores as a bimodal Gaussian mixture model (GMM) to produce GMM masks. Additionally, we remove orphan pixels by computing the mean filter over the resulting binary classification map to produce our preliminary masks M_{pre}. These preliminary masks could potentially already be used to detect objects as evaluated in Sec. 4.

Fine-tuning and Mask Refinement: The main problem with the preliminary masks is that, because they are derived from the latent space of the LDM, their resolution is only 64×64. This limits them to being a coarse segmentation, which often overestimates the size of the object rather than following the object's sharp edges. A rough segmentation is sufficient for tasks like inpainting, especially if the area around the object is homogeneous, but for the task of extracting the foreground object it produces unwanted artefacts. To work around this, we leverage the same diffusion model that we used to extract the masks. We use the binary classification prediction of the GMM mask M_{pre} to select random rectangles within the image that only contain background. Then we fine-tune the model to inpaint only the background of these images by conditioning on the prompt y_b =“a photo of a background” using the image as ground truth (See supplements for examples). Simultaneously, the model is fine-tuned on the task of full image synthesis to generate new samples for $\mathbf{D}'$ by conditioning on y_f = “a photo of a {object}”.

To generate the final foreground masks $\mathbf{D_m}$ we use the fine-tuned LDM m' to inpaint the background over the area covered by the preliminary mask conditioning on the background prompt. To identify the foreground region, we take the pixel-wise intensity difference between the background-inpainted and original images. As the inpainted images are conditioned to generate background, the difference is higher in the true foreground region. We apply a Gaussian mixture model to the pixel-wise difference map to create a binary classification map. Formally the refined masks M are computed as

$$M = M_{pre,up} \odot g(|\mathbf{x} - m'(\tilde{\mathbf{z_0}}, y_b)|), \quad (5)$$

$$\tilde{\mathbf{z_0}} = \mathbf{z_0} \odot (1 - M_{pre}) + \mathbf{z} \odot M_{pre} \quad (6)$$

where $M_{pre,up}$ denotes the preliminary mask upsampled to the input image's resolution, g applies the bimodal Gaussian mixture model to the pixel values and $\tilde{\mathbf{z_0}}$ is the latent code of an image with the foreground region replaced with random noise $\mathbf{z}$. This produces refined masks, as the use of pixel-wise error improves the segmentation around the sharp edges of the object. Additionally, the computation of the refined masks is performed in pixel-space of the images instead of the latent space of m and therefore produces even more detailed masks.

Finally, to further improve our mask prediction we use the refined masks as labels to train a U-Net [45] to directly segment the foreground of the images, similar to the approach in [61], which follows a standardized approach of training a U-Net on a fixed number of steps and hence does not require any hyperparameter tuning. We experiment with training the segmentation network on refined masks of the original unlabelled training data ($\mathbf{D_s}$), as well as training it with the fully synthetic dataset $\mathbf{D'_s}$ as an augmentation method, which we generate by prompting the fine-tuned model with the foreground conditioning y_f and repeating our pipeline of mask refinement on this synthetic dataset to get segmentation labels.

4. Evaluation and Results

Implementation: We use PyTorch 1.11 and run our experiments on a workstation with two A6000 Nvidia GPUs. Concept distillation training takes on average one day. The forward pass is fast, equivalent to that of a standard U-Net.

Datasets: We choose our datasets such that they cover a variety of different objects, including in-the-wild animals and cars, as well as humans in static setups.

Human3.6m [24] is a dataset of 3.6 million images of humans in different scenarios and situations. To show that our method does not rely on large datasets we take a subset of 6000 randomly chosen images centred around the human and cropped to 256×256 pixels from the training dataset of Human3.6m.

To test the method on representations of animals, we use two datasets: the Stanford Dog Dataset [28], which contains 20,580 images of dogs divided into different categories, and the Caltech-UCSD Birds 200 (CUB) dataset [58], which contains 11,788 images of birds from 200 different species.

Finally, we also experiment with the detection of cars using [30], which consists of 16,185 images of cars in different natural and non-natural settings. All these datasets come with subcategories grouping the images based on selected features. For our use case, we consolidate these groups when prompting the model and use the classes *cars*, *dogs*, *human*, and *birds* to simulate the absence of manual labels.

Methods	CUB		
	ACC	IoU	mIoU
Fully supervised U-Net	97.9	88.3	93.0
GrabCut [46] by [49]	72.3	36.0	52.3
ReDO [11]	84.5	42.6	–
PerturbGAN [4]	–	–	38.0
IEM + SegNet [49]	89.3	55.1	71.4
Melas-Kyriazi et al. [35]	92.1	66.4	–
Layered GAN [61]	94.3	69.7	81.7
Ours (U-Net trained on $\mathbf{D_s}$)	95.2	75.1	84.8
Ours (U-Net trained on $\mathbf{D_s} \cup \mathbf{D'_s}$)	**95.6**	**77.2**	**86.0**

Table 1. Comparison to other segmentation methods. Baselines taken from [61]. Details on the training of the U-Nets can be found in the appendix.

Self-supervised Segmentation Performance: Table 1 compares our methods performance against other unsupervised methods on the CUB dataset, showing the pixel-wise accuracy, the Intersection over Union (IoU) of the foreground segmentation and the mean IoU. From this we see that training a U-Net on our self-supervised labels produces a model that outperforms all other methods, achieving an overall foreground IoU improvement of 5.4 compared to [61]. Furthermore, by adding the fully synthetic dataset $\mathbf{D'_s}$ to the training data we are able to improve the performance even further, reaching a foreground IoU of 77.2. Table 3 also shows high foreground IoU values across the other datasets, with Fig. 4 showing qualitative examples.

Mean Attention Map Performance: Computing the mean attention maps $\hat{M}$ and comparing them to the ground-truth yields a remarkable AUC-ROC for the bird dataset of 97.1. Scores are normalized instance-wise to a range of 0 to 1. We also experiment with no normalization, which gives a slightly worse AUC-ROC of 97.08. Qualitative examples from all datasets are shown in Fig. 5, displaying the mean attention maps' ability to localise the foreground in different scenarios. To compare this to our classification results we compute the threshold such that we reach over 95% true positive rate on a reserved training set of 100 images. True positive rate is more important in our case because we observe that falsely classifying pixels as background leads the inpainting model taking these foreground pixels as sources to inpaint larger parts of the image. We reach a pixel-wise accuracy of 86% on a set of 1000 test images suggesting that our method of extracting the classification masks already can provide meaningful results.

While these results are encouraging, they also require a ground-truth dataset and thresholding that we do not want to rely on. The results indicate that our computed masks

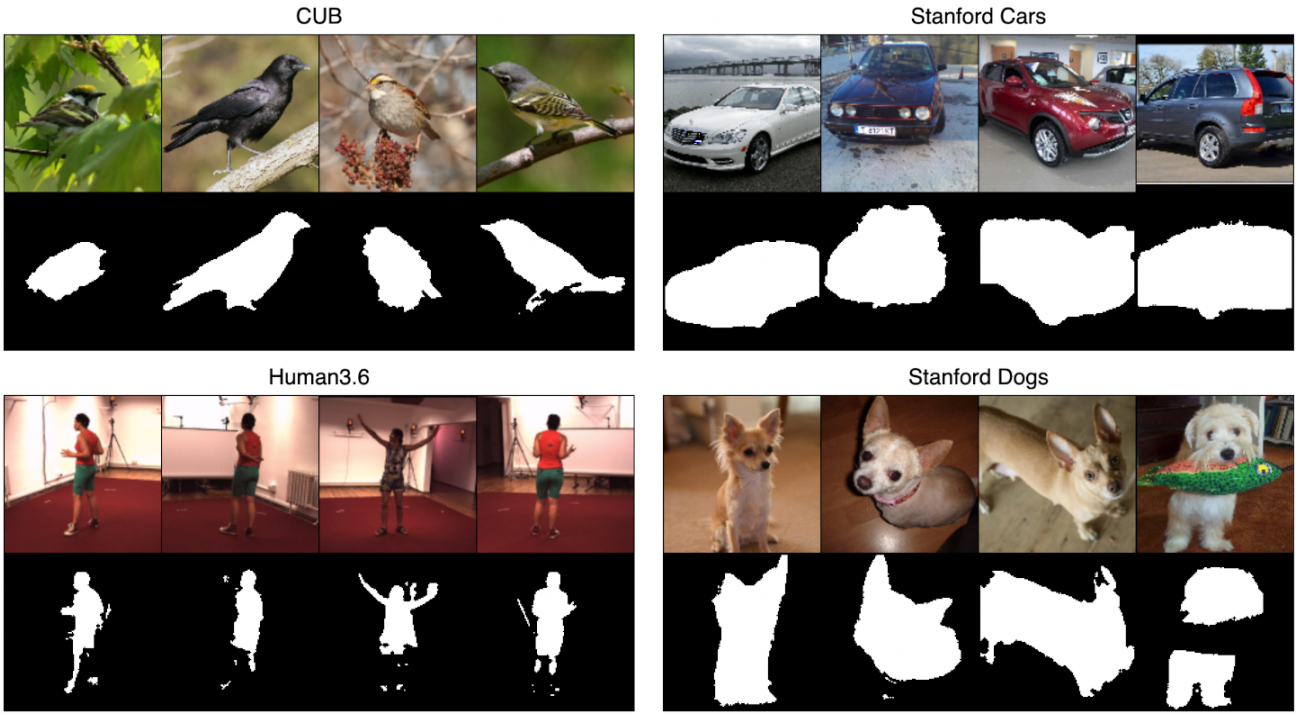


Figure 4. Unsupervised Segmentation Masks generated by our proposed approach.

are very good at locating the objects, albeit they do not reach state-of-the-art performance despite adding supervision. Training on refined masks and integrating synthetic data surpasses supervised results without using any labels as shown in the next section.

Methods	CUB		
	ACC	IoU	mIoU
Preliminary Masks M_{pre}	83.5	29.0	55.7
Simple Inpainting M_{crop}	90.0	30.7	60.0
U-Net trained on ($\mathbf{D}$, M_{pre})	91.7	66.5	78.2
Refined Masks M	92.4	63.6	77.4
U-Net trained on $\mathbf{D_s}$	95.2	75.1	84.8
U-Net trained on $\mathbf{D_s} \cup \mathbf{D'_s}$	**95.6**	**77.2**	**86.0**

Table 2. Segmentation results for different steps of our pipeline.

Ablation study: Table 2 shows an ablation study for the individual components of our method. Initially, the zero-shot preliminary masks generated from the foundation model achieve good accuracy (83.5), but poor foreground IoU (29.0). This reflects our earlier intuition that masks overestimate the size of the object due to the maps being computed at a lower resolution. Thanks to this, the GMM overestimates the boundaries and spares the need for any hyperparameter. Optimizing the threshold to maximise the accuracy over a set of 100 training images would increase the accuracy to 93.9%, but requires a ground-truth dataset.

Training a U-Net on these labels (M_{pre}) increases the IoU to 66.5, showing the value in training the segmentation network. Using the refined masks as foreground segmentation gives comparable performance, with higher accuracy but lower foreground IoU. Additionally, we experiment with replacing the inpainting step used to improve M_{pre} with a simpler approach that crops background areas and uses them as inpainting in Eq. (5) instead of m'. The resulting masks M_{crop} are worse than the masks from our proposed refinement step (For more details see supplements). However, training a U-Net on these refined masks produces the best results, being further boosted by incorporating additional synthetic data. The progressive refinement of the segmentation masks is shown in Fig. 6.

Data synthesis: Table 3 compares the generative ability of our fine-tuned diffusion model using the Fréchet inception distance (FID). Our method achieves higher generation quality than all other methods across the CUB, Stanford Dogs, and Stanford Cars datasets. We improve upon LayeredGAN's remarkably low FID scores for CUB and Stanford Cars by 3.1 and 5.6 respectively.

Our method allows for the generation of samples covered entirely by the background, as shown in Fig. 7, without ever seeing such an image during training. The success of this component is what enables the refined masks to be generated, as accurately inpainting the foreground allows us to use the pixel-wise difference between the original and inpainted images to precisely identify the foreground.

Concept Distillation: Finally we can evaluate if our model

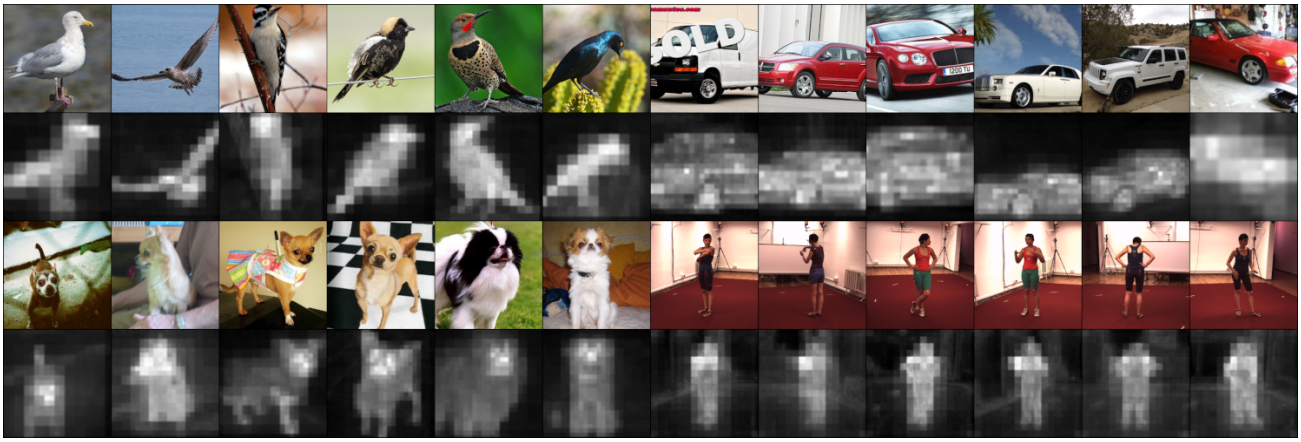


Figure 5. Mean attention maps for all datasets in latent space $\mathbf{z}$ of the diffusion model. Prompts are "a photo of a {object}", where {object} is replaced by "bird" for the first pair of rows, then "car", "dog", and "human".

Methods	Sup.	CUB		Stanford Dogs		Stanford Cars		Human3.6m	
		FID ↓	IoU ↑	FID ↓	IoU ↑	FID ↓	IoU ↑	FID ↓	IoU ↑
FineGAN [54]	Weak	23.0	44.5	54.9	-	24.8	-	-	-
OneGAN [3]	Weak	20.5	55.5	48.7	-	24.2	-	-	-
LayeredGAN [61]	Unsup.	12.9	69.7	59.3	-	19.0	-	-	-
Ours	Self	**9.8**	**75.1**	**43.1**	**63.8**	**13.4**	**55.2**	**63.7**	**69.2**

Table 3. Quantitative Results on $\mathbf{D_s}$. Training details are shown in the supplements. Values are taken from [61]. Source code for [54, 3, 61] was not available for re-evaluation on the Dogs, Cars and Human3.6m datasets. IoU on CUB are reported using the prediction of our model and the ground truth provided with the dataset. For the other datsets we use the IoU of the bounding boxes.

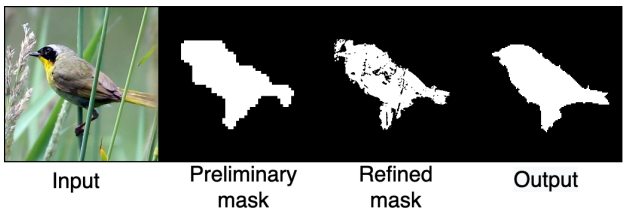


Figure 6. Progressive refinement of the segmentation masks.

has indeed learned to distinguish between foreground and background by looking at the output of different classifier-free guidance scales starting from the same seed $\mathbf{z_t}$. Since we have fine-tuned our model only on two distinct textual prompts the conditional image generation should have collapsed to two clusters, one for the object and the other one for the background. Hence, instead of performing classifier-free guidance using the predictions of m' conditioned on empty prompts we can directly use the predictions of $m'(\mathbf{z_t}, y_b)$ and $m'(\mathbf{z_t}, y_f)$ to perform image interpolation. We define classifier-free guidance in the direction of the foreground, hence, a higher score means that latent representations are pushed further in the direction of the object. The results are shown in Fig. 8. For negative guidance scales, the background is more detailed and there is no bird present. Increasing this scale leads to less detail in the background while birds often seem to naturally grow from the details in the background. The quality of the birds visually improves while the quality of the rest of the image keeps degrading resulting in less detailed backgrounds. We confirm this quantitatively by computing the FID for different classifier-free guidance scales. Without classifier-free guidance, *i.e.*, scale = 1, the method reaches a FID of 9.8, at scale = 3 a FID of 11.3, and at scale = 7.5 a FID of 22.3.

Medical Image Analysis: We want to evaluate if this approach can be applied to other domains, such as medical imaging. Since the LDM does not have any medical understanding, we first need to fine-tune it using MIMIC [27], which provides chest x-ray images paired with radiology reports. We can fine-tune the model using a similar approach as the one suggested by [8] (Details on the fine-tuning can be found in the supplements). Then we report the pixel-wise AUC-ROC on MS-CXR [5], a subset of MIMIC with bounding box labels for diseases. Qualitative results can be seen in Fig. 9. The pixel-wise accuracy of the attention mask is already at 79.6% AUC-ROC across eight different diseases, however, the bimodal GMM assumption no longer holds in many cases because the model distinguishes three regions, namely: background, foreground, and the rest of the chest region.



Figure 7. Full background image synthesis from the fine-tuned model, conditioned on y_b. Using our proposed fine-tuning method, the diffusion model is successfully able to generate images without birds from a dataset only consisting of images with birds.

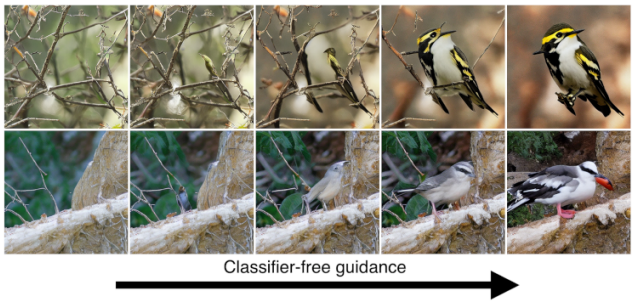


Figure 8. Synthetic results of m' with changing scales of classifier-free-guidance, ranging from -2 on the left to +7.5 on the right.

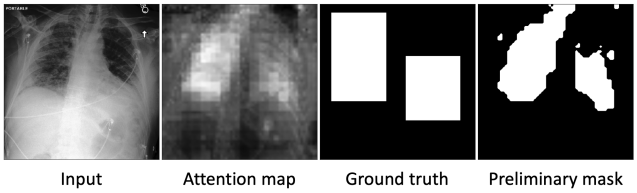


Figure 9. $\hat{M}$ and M_{pre} extraction on a medical task.

5. Discussion

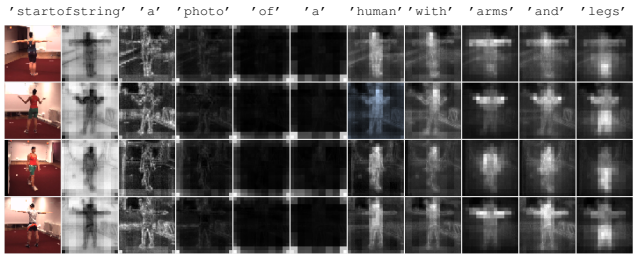


Figure 10. Input image and mean attention maps for every word of the textual input prompt "startofstring a photo of a human with arms and legs". The attention of "human" focuses on the torso, the one for the "arms" on the arms, and the one for the "legs" on the legs.

We show in Sec. 4 that our method to extract the segmentation masks M from the preliminary attention masks yields better results than computing an optimised threshold over a reserved miniset. This is possible because comparing to the inpainted background provides sharp edges around the object. However, our model is currently limited to detecting single object concepts. An extension to multiple objects could be achieved by prompt engineering in combination with data augmentation techniques. Taking the foreground masks and using them to extract objects would enable multi-instance and multi-object segmentation by layering multiple objects over each other and extending the final U-Net to a multi-label segmentation model.

Furthermore, learning from weak labels has the disadvantage that the segmentation model may learn and reproduce weaknesses of the initial method. In our case, the bimodal GMM fails if the image has more than two distinct contrast clusters. This is especially the case on the Human3.6m dataset where the floor, walls, and person have vastly different contrast-levels. Consequently, the final segmentation sometimes fails to detect the lower part of the body as shown in Fig. 4. However, our method could easily be adapted to only a part of the human body, such as the legs. We show this in Fig. 10, as by conditioning images from [24] on the prompt "*A photo of a human with arms and legs*" and computing $\hat{M}$ for the three concepts (human, arms, and legs) we are able to produce attention maps focused on specific body parts.

6. Conclusion

In this work, we have presented a generalizable framework to train segmentation networks without any hyperparameter tuning using an unsupervised zero-shot approach following the paradigm of 'textual concept description' $\rightarrow$ 'segmentation model'. We leverage the power of large generative latent diffusion models and fine-tune the model on the task of generating foreground and background images, which can be used as data augmentation methods. We show, that this method can achieve results close to supervised methods, without requiring any manually generated ground

truth labels. Our approach is amenable to supervised deep learning and can be combined with existing models to boost segmentation performance even further.

In future work we will explore how multi-object, multi-instance segmentation can be facilitated with concept distillation from generative image foundation models.

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